

Prompt: write a program for c that chooses a number from one to ten, then allows the user to try and guess that number three times. let the user know if their guess is too high or low, and if it is correct, end the loop immediately. give the user a final win or lose message depending on if they guessed correctly.

Gemini Code (prompted online on personal computer (gemini pro)):

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main() {
    int secret_number, guess;
    int max_attempts = 3;
    int win = 0;

    // Seed the random number generator using the current time
    srand(time(NULL));

    // Generate a random number between 1 and 10
    secret_number = (rand() % 10) + 1;

    printf("I have chosen a number between 1 and 10.\n");
    printf("You have %d attempts to guess it!\n\n", max_attempts);

    // Loop for the allowed number of attempts
    for (int attempt = 1; attempt <= max_attempts; attempt++) {
        printf("Attempt %d: Enter your guess: ", attempt);
        scanf("%d", &guess);

        // Check the user's guess
        if (guess == secret_number) {
            win = 1;
            break; // End the loop immediately if correct
        } else if (guess < secret_number) {
            printf("Too low!\n\n");
        } else {
            printf("Too high!\n\n");
        }
    }

    // Print final win or lose message
    if (win) {
        printf("Congratulations! You guessed the correct number: %d\n", secret_number);
    }
}
```

```

    } else {
        printf("Game over! You've run out of attempts. The correct number was: %d\n",
secret_number);
    }

    return 0;
}

```

Copilot's code (also prompted online on computer):

```

#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main(void) {
    srand(time(NULL));          // seed the random number generator
    int secret = rand() % 10 + 1; // random number from 1 to 10
    int guess;
    int won = 0;

    for (int i = 1; i <= 3; i++) {
        printf("Attempt %d: Enter your guess (1–10): ", i);
        scanf("%d", &guess);

        if (guess == secret) {
            printf("Correct! You guessed the number.\n");
            won = 1;
            break;
        } else if (guess < secret) {
            printf("Too low.\n");
        } else {
            printf("Too high.\n");
        }
    }

    if (won) {
        printf("You win!\n");
    } else {
        printf("You lose. The number was %d.\n", secret);
    }

    return 0;
}

```

COMPARISON:

After looking through the codes given, both seem very similar, with differences only really being where they are placed. Gemini's code includes more comments and has wordier print statements, as well as an extra variable that holds the max attempts. While it is unnecessary because the max number should always be 3, i believe that for maintainability purposes it might be better to have a variable control that, in case i ever wanted or needed to change the amount of guesses. Thus, i will be moving on with Gemini's code.

- Modifying code

Thinking about bare necessities, what is most important for this code is a random number, a for loop for the number of attempts, and if conditionals that check the input from the user. Because this was mainly what the code was, managing the execution time and space complexity may not exactly be possible any further, at least to what i can see.

(I did want to look into potentially getting rid of the if statement from the win condition and making it like a for-else loop, but i believe that may be more complicated as it is not a built-in function like python is)

As for maintainability, i went ahead to flesh out the comments a little more.

(Touched-up code)

```
//Output:  
//No collaborators  
//Code Generated with Gemini, No code used from Copilot generation with  
same prompt, no other sources used  
//Name: Christopher Warren
```

```
#include <stdio.h>  
#include <stdlib.h>  
#include <time.h>  
  
int main() {  
    int secret_number, guess;  
    int max_attempts = 3;  
    int win = 0;  
  
    // Seed the random number generator using the current time  
    srand(time(NULL));  
  
    // Generate a random number between 1 and 10
```

```

secret_number = (rand() % 10) + 1;

printf("Guess a number between 1 and 10.\n");

// Loop for the allowed number of attempts,
// allowing the user to guess until they guess correctly or until
theyre out of attempts.
//Lets the user know if too high or too low.
for (int attempt = 1; attempt <= max_attempts; attempt++) {
    //tells user instructions, amount of guesses
    printf("Attempt %d: Enter your guess: ", attempt);
    //user input
    scanf("%d", &guess);

    // Check the user's guess.
    if (guess == secret_number) {
        win = 1;
        break; // set win to true and end the loop immediately if
correct
        } else if (guess < secret_number)    { //tells the user if too high
or too low
        printf("Too low!\n\n");
        } else {
        printf("Too high!\n\n");
        }
    }

// Print final win or lose message (based on win variable)
if (win) {
    printf("Congratulations! You guessed the correct number.");
} else {
    printf("Game over! You've run out of attempts. The correct number
was: %d\n", secret_number);
}
// end
return 0;
}

```

